# Utility Analysis Dashboard

Technical Design Note

Generated 2026-03-10 · Internal reference — do not distribute

## 1. Stack

**Runtime:** Python 3.11+, Streamlit, Plotly (go + px), pandas / numpy

**Venv:** shared `../finance_vis/venv_finance` — do not create a new one

**Launch:**

```
source ../finance_vis/venv_finance/bin/activate && streamlit run app.py
```

**Input:** Korean `.xlsm/.xlsx` files loaded from `./data/` (configurable path in sidebar)

## 2. Module Map

```
app.py              ← Page config, top-level routing, file loading
sidebar.py          ← File loader (dir scan → _FileEntry list), bins/tail stubs
data.py             ← Excel I/O (@st.cache_data), sheet name constants, billing period detection
features.py         ← Column engineering: create_change_columns, build_from_two_files,
                       aggregate_by_brand, split_brand_by_floor, floor parsing helpers
filters.py          ← render_meter_filters, show_filter_widgets, apply_sheet_filter,
                       brand_search_bar
viz.py              ← plot_hist_with_tails (Plotly go.Bar histogram with IQR/tail shading)
lang.py             ← t() translation helper (Korean only, key→string)

meter_view.py       ← ■■■■ full pipeline: load → filter → histograms → 5 tabs
summary.py          ← ■ ■■■■ ■■ + ■■■ ■■■ (water/hotwater/electricity summary)
billing.py          ← ■■■■■ ■■ ■■ view
ehp.py              ← EHP(OAC)■■■■ view
water.py / hotwater.py / electricity.py  ← per-sheet views
brand_profile.py    ← Brand profile tab with peer comparison charts

tab_anomaly.py      ← ■■■■ tab (composite score, heatmap, spike/cost/HVAC sub-tabs)
tab_cross.py        ← ■■■■ tab (unit costs, electricity breakdown)
tab_efficiency.py   ← ■■■■ tab (per-m² usage benchmarking)
tab_corr.py / tab_reconciliation.py  ← correlation & reconciliation tabs

anomaly_features.py ← build_anomaly_df: merges meter+billing+elec+water → anomaly scores
cross_features.py   ← build_unit_costs, build_elec_breakdown
report.py           ← generate_report_pdf (■■■■ main report)
biz_report.py       ← generate_anomaly_pdf, generate_cross_pdf, generate_efficiency_pdf
billing_report.py / ehp_report.py / hvac_report.py  ← per-sheet PDF generators
```

## 3. Navigation Architecture

```
sac.tabs (sidebar, left-position) → nav_mode
  ■■■ ■■ ■■        → sheet selectbox → billing/ehp/water/hotwater/electricity/meter_view
  ■■■ ■■             → analysis selectbox
  ■     ■■■ ■■ ■■  → render_summary_view(water_df, hw_df, el_df)
  ■     ■■■ ■■ ■■  → render_meter_filters → [■■■■|■■■■|■■■■] tabs
  ■■■ ■■■ ■■■     → render_brand_profile_tab
```

**Key quirk:** `sac.tabs` must render directly inside `with st.sidebar:` — **never** inside `st.empty()`. The empty placeholder causes the custom component to re-initialize on every rerun, losing state.

## 4. Data Pipeline (■■■■)

```
read_sheet()                        # raw Excel, header=[2, 3, 4]
  ■■ apply_header_rows()            # flatten MultiIndex headers → named columns
       ■■ build_from_two_files()  # prev file → *_previous = prev month usage
            ■■ create_change_columns()  # *_change = curr - prev, *_pct = change/prev*100
                 ■■ aggregate_by_brand()    # group by (brand, building), sum usage
                      ■■ split_brand_by_floor()  # if floors filtered, divide totals equally
```

**Cumulative vs usage:** Meter readings in the Excel are cumulative. `build_from_two_files` renames them to `*_meter_curr/prev` and uses the pre-computed usage column (`water_usage_m3` etc.) as the actual current-month consumption.

### *Column Naming Convention*

| Column | Meaning |
|---|---|
| water_previous / water_current | This month's usage from prev/cur file |
| water_change | current – previous |
| water_pct | change / previous × 100 |
| water_meter_prev / water_meter_curr | Original cumulative readings (kept for backward detection) |

## 5. Floor Logic

Floor values are compound strings: `"1F/2F"`, `"2F~5F"`, `"B2F/B1F"`.

| Function | Purpose |
|---|---|
| parse_floor_value(s) | Returns list of individual floor strings from compound value |
| get_simple_floors(df) | Sorted unique floors for multiselect widget |
| split_brand_by_floor(df, sel_floors) | Divides brand totals equally by count of matched floors |

When `sel_floors == ["All"]` and `sel_bldg == ["All"]` → no splitting, use aggregate totals.

## 6. Histogram System (viz.py: plot_hist_with_tails)

### *Signature (key params)*

```
plot_hist_with_tails(s, bins, lo, hi, title,
    source_df=None, val_col=None, key="hist",
    display_cols=None, tail_pct=None, val_scale=1.0)
```

• **lo / hi** are in the **same units as the plotted series s**

• **val_scale** bridges the gap when s is scaled (e.g. `/1e4` for ■■) but `source_df[val_col]` is raw ■:

```
mask = (source_df[val_col] / val_scale >= x0) & (source_df[val_col] / val_scale <= x1)
```

• Style is **locked**: blue normal bars (#4C72B0), amber tail bars (#DD8A00), red median line (#C44E52), white bg, height=380px

### *IQR Detection Pattern*

```
q1, q3 = s.quantile(0.25), s.quantile(0.75)
iqr    = q3 - q1
lo     = q1 - k * iqr   # k slider: 0.5–3.0, default 1.5, step 0.25
```

```
hi     = q3 + k * iqr
# Show LaTeX:
st.markdown(
    f"$$Q_1={q1:,.0f},\;Q_3={q3:,.0f},\;IQR={iqr:,.0f}$$\n\n"
    f"$$\\text{{Lower}}={lo:,.0f},\;\\text{{Upper}}={hi:,.0f}\;(k={k})$$"
)
```

## 7. Money Formatting

```
def _fmt_won(v):
    if abs(v) >= 1e8: return f"{v/1e8:,.0f} ■■"
    if abs(v) >= 1e4: return f"{v/1e4:,.0f} ■■"
    return f"{v:,.0f} ■"

# For bar charts — compute _div from max(series):
_div, _unit = (1e8, "■■") if _max >= 1e8 else (1e4, "■■")
_xv = series / _div
# pass val_scale=_div to plot_hist_with_tails so bin-click filter still works
```

Rule: **no decimals on ■ values** — always `:.0f`.

## 8. Anomaly Scoring (anomaly_features.py)

Composite score = weighted sum of 5 components, each normalized to [0, 1]:

| Component | Weight | Signal | Source sheet |
|---|---|---|---|
| ■■ Spike | 30% | MoM % change vs thresholds (100/50/20%) | ■■■■ |
| ■■ Consumption | 25% | Quadrant scores: HH=4, HL=3, LH=2, Normal=1, LL=0 | ■■■■ |
| ■■ Cost | 25% | Z-scores of ■/m³, ■/kWh, ■■/m² | ■■■■■ ■■ ■■ |
| HVAC | 10% | kWh/m² IQR-normalized | ■■ ■■ ■■■■ |
| ■■■ Consistency | 10% | Count of zero-usage utilities | ■■■■ + sheets |

**Risk levels:** ■ ■■ ≥ 0.65 · ■ ■■ ≥ 0.40 · ■ ■■ ≥ 0.20 · ■ ■■ < 0.20

## 9. PDF Generation Pattern (all biz tabs)

```
_pdf_key = f"{tab}_pdf_{file_name}"
_col_gen, _col_dl = st.columns([1, 2])
with _col_gen:
    if st.button("■ PDF ■■■ ■■", key=f"gen_{tab}_pdf_{file_name}"):
        with st.spinner("PDF ■■ ■…"):
            st.session_state[_pdf_key] = generate_X_pdf(df)
if _pdf_key in st.session_state:
    with _col_dl:
        st.download_button("■■ ■■■■", st.session_state[_pdf_key], ...)
```

Generate-then-cache pattern: PDF bytes stored in session_state to avoid re-generating on every Streamlit rerun. Generate button and download button are separate widgets in adjacent columns.

## 10. Common Pitfalls

| # | Issue | Fix |
|---|---|---|
| 1 | Closure bug in loops | Use default arg binding: def fn(key, _p=p, _cc=cc): — not bare p/cc |

| # | Issue | Fix |
|---|---|---|
| 2 | sac.tabs state loss | Never wrap in st.empty(). Use with st.sidebar: directly. |
| 3 | Bin-click scale mismatch | When series is scaled before histogram, pass val_scale=scale_factor to plot_hist_with_tails |
| 4 | Radio option ordering | Histogram first — ["[redacted]", ...] or ["% Change only", "Change only", "Side by Side"] |
| 5 | Decimal on [redacted] values | Always :.0f, never :.2f |
| 6 | EHP column parsing | Cumulative cols M–DG (0-based 12–110); stop at next [redacted] header; do NOT merge two tables |
| 7 | sac.tabs ValueError | Raises if stored index ≥ number of tabs. Clear session state key nav_{file_name} to reset |

## 11. Notable Session State Keys

| Key | Purpose |
|---|---|
| nav_{file_name} | Sidebar nav tab index (sac.tabs) |
| anomaly_loaded_{file_name} | Gate: has anomaly analysis been started? |
| cross_loaded_{file_name} | Gate: has cross-sheet data been loaded? |
| anomaly_pdf_{file_name} | Cached anomaly PDF bytes |
| cross_pdf_{file_name} | Cached cross-tab PDF bytes |
| efficiency_pdf_{file_name} | Cached efficiency PDF bytes |
| {prefix}_iqr_k | IQR k multiplier per histogram |
| {prefix}_bins / {prefix}_bins_i | Bins slider/input sync pair |

## 12. Pending / Known Issues

- [redacted] **unit scaling**: Should display as [redacted]/[redacted] (not bare [redacted] or [redacted]/$m^2$). Implementation was reverted at user request — ready to re-implement cleanly.
- No other explicit outstanding tasks as of 2026-03-10.